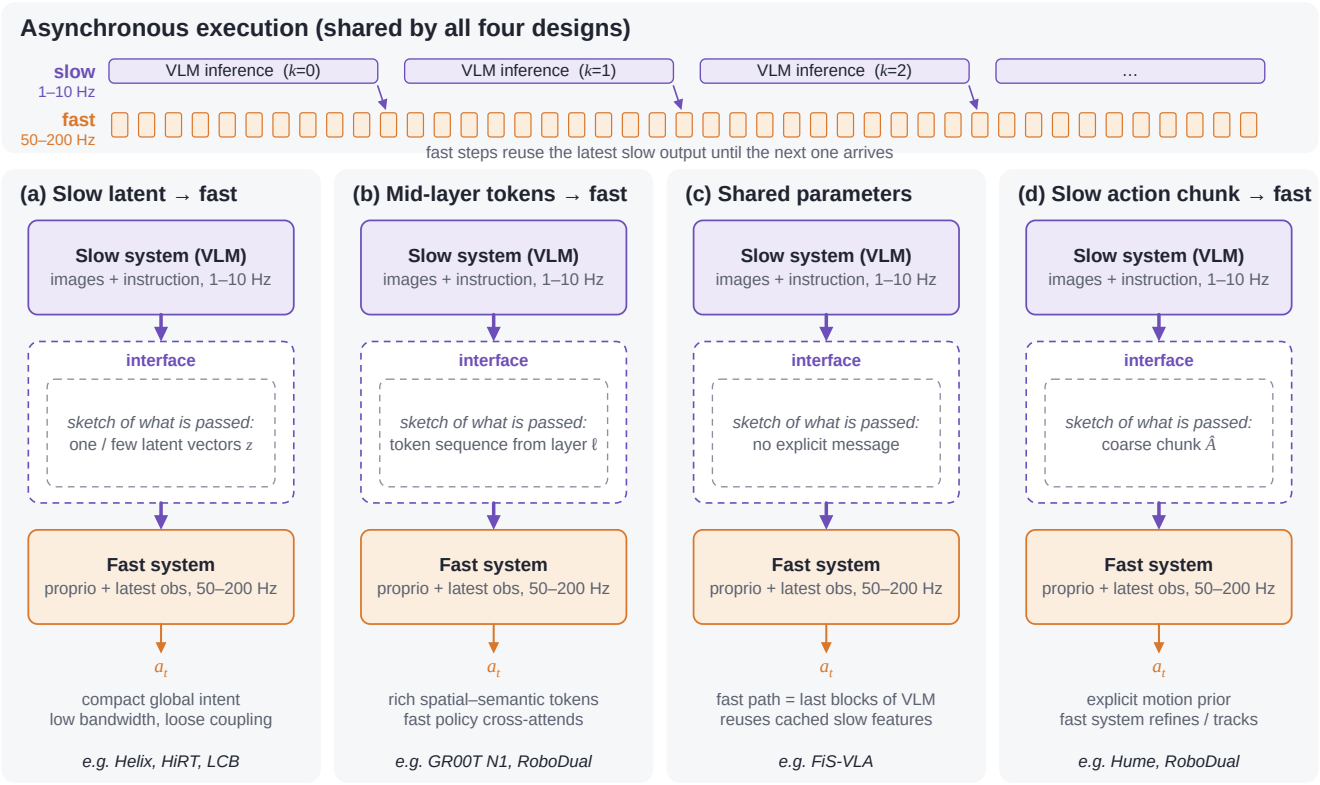

Asynchronous execution (shared by all four designs)
slow
1–10 Hz
VLM inference (k=0)
VLM inference (k=1)
VLM inference (k=2)
…
fast
50–200 Hz
fast steps reuse the latest slow output until the next one arrives
(a) Slow latent → fast
Slow system (VLM)
images + instruction, 1–10 Hz
interface
sketch of what is passed:
one / few latent vectors z
Fast system
proprio + latest obs, 50–200 Hz
a_t
compact global intent
low bandwidth, loose coupling
e.g. Helix, HiRT, LCB
(b) Mid-layer tokens → fast
Slow system (VLM)
images + instruction, 1–10 Hz
interface
sketch of what is passed:
token sequence from layer ℓ
Fast system
proprio + latest obs, 50–200 Hz
a_t
rich spatial–semantic tokens
fast policy cross-attends
e.g. GR00T N1, RoboDual
(c) Shared parameters
Slow system (VLM)
images + instruction, 1–10 Hz
interface
sketch of what is passed:
no explicit message
Fast system
proprio + latest obs, 50–200 Hz
a_t
fast path = last blocks of VLM
reuses cached slow features
e.g. FiS-VLA
(d) Slow action chunk → fast
Slow system (VLM)
images + instruction, 1–10 Hz
interface
sketch of what is passed:
coarse chunk $\hat{A}$
Fast system
proprio + latest obs, 50–200 Hz
a_t
explicit motion prior
fast system refines / tracks
e.g. Hume, RoboDual